Pole Unit - Electronics and Data Flow

Hardware interfaces | Edge processing | Firmware authorization | Independent power

V-ORRA | Intelligent Vehicle Signalling & Connected Road Safety System | VK26-IITH-SC-0177 | Vishwakarma Awards 2026-27

ITM University Gwalior | 27 October 2026 preparation

ROADSIDE POLE UNIT

E Electronic controllers | F Firmware & logic | M Structure | A Power distribution

UWB Two-Way Ranging - IEEE 802.15.4z-compatible hardware

Texas Instruments AWRL6432BOOST

60 GHz FMCW; range / angle / Doppler / detections
J5 Micro-USB: XDS110 user UART and power
Output format depends on chosen radar firmware

USB 2.0
XDS110
Data + Power

Makerfabs MAUWBS3CA1 - UWB Anchor

STM32 / DW3000; onboard ESP32-S3 bridge
USB-UART reports: tag ID / range / time / quality*
Anchor report bridge and firmware to be verified

USB Serial
Range reports

Powered USB Hub

4-Port USB 3.0 Hub

P1 - Radar J5 Micro-USB

P2 - Anchor USB-UART port

P3 - Pole safety-gate USB serial

P4 - Reserved

USB-A upstream to Pi USB-A host
Separate 5 V auxiliary input

USB-A upstream
USB host

Raspberry Pi 5 - Edge Processing Unit

Model B, 1 GB | Official Active Cooler to fan connector

P1 / P2 Radar + UWB Acquisition

USB-host device streams; firmware-specific parsers
Retain timestamps and validity / quality where available

Aligned measurements

P3 Calibration & Time Association

Radar coordinate transform; sensor mounting offsets
Timestamp alignment within available timing accuracy

Calibrated observations

P4 Vehicle Tracking & Sensor Association

Multi-frame radar tracks vs UWB range / tag evidence
Motion consistency; ambiguity detection; no identity from radar

Track / association / confidence

P5 Confidence-Aware Geofence Decision

Associated track + surveyed boundaries + uncertainty
Inside NHZ / LBZ | Outside | UNKNOWN

Provisional zone state

P6 Vehicle-Specific Command Proposal

Provisional state / vehicle / session / sequence / expiry
USB host -> hub P3 -> separate pole gate

Configuration Store

Surveyed road / NHZ / LBZ
Approved version / policies
Logical authorized deployment

Events

P8 Logging / Diagnostics

Radar / range observations
Tracking and association
Zone and proposal events
Gate outcomes; ACK; faults

Side branch only

ACK / faults

P7 Local Network

Pi 2.4 GHz Wi-Fi AP
Mosquitto MQTT broker

Gate + vehicle: Wi-Fi clients
All MQTT through this broker
No Pi bypass of gate approval

Authenticated
vehicle command

Applied-State ACK
+ Fault Status

USB Serial
Proposals / Diagnostics

Physical ARM / DENY Input

ARM/DENY input; switch to local ground
ARM = LOW; DENY = HIGH
Startup defaults to DENY
Gate status LED

ARM / DENY

ESP32-S3 - Pole Safety Gate

USB serial: proposals and diagnostics
Approve only when ALL pass:
ARM enabled; proposal valid
Association acceptable; confidence sufficient
Session valid; command fresh
Watchdog / authorization healthy

Otherwise: deny new restrictive commands
Firmware-controlled authorization only
No safety certification / power disconnect

Logical proposal
via USB host
/ hub P3

M Mechanical Structure & Mounting

Approximately 2 m circular vertical support
Top-mounted electronics enclosure / sensors
Adjustable radar and UWB mounting brackets
Weighted base, structural support and cable routing
Alignment offsets feed calibration parameters

2.4 GHz Wi-Fi / MQTT - gate-authorized publication; ACK / status reception

A POLE POWER DISTRIBUTION - appliance-level AC connections; separate regulated supplies

100 W / 156 Wh Power Station
Commercial AC output

AC

Rated AC Extension Strip
Purchased appliance wiring

Official Raspberry Pi 27 W USB-C Supply → **5.1 V DC / 5 A -> Pi USB-C | Cooler -> official fan connector**

Regulated Adapter → **Regulated DC -> Powered USB Hub AUX**

Remote Makerfabs bare MaUWB tag

Ranging only; no actuation command transport
Single anchor supplies a distance constraint
Vehicle uses an independent demo power supply

Remote Vehicle ESP32-S3

Wi-Fi / MQTT client; subscribed vehicle identity
Independently validates gate authorization
Applied-state ACK + fault-status publication
Designed authentication / replay / expiry checks

INTERFACE / VALIDATION NOTES

* Time / quality fields depend on firmware.
Radar data format and host parsing to be fixed.
Anchor USB bridge behavior to be verified.
Hardware capability does not prove secure ranging.
Hub distribution limits / backfeed unverified.
Kalman / assignment methods are candidates.
Prototype physical validation is in progress.
Auxiliary demo only; closed-course only.
Local grounds; no real-vehicle horn/headlamp wiring.
No pole-to-vehicle cable; independent demo supplies.

DECISION RULE | UNKNOWN or ambiguous association cannot authorize a new restriction.
Security is an application-layer design requirement; connectivity alone never authorizes actuation.

LINE CONVENTIONS | Arrow direction indicates sender to receiver; matching named power nets are continuous locally.

Wired / logical data | Wireless RF | Power / voltage | Configuration | Inhibit / output control

M Mechanical structure | A Actuation & power | E Electronic controllers | F Firmware & logic | Dashed box: optional feature